AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 14 · STUDENT LESSON

Statistical tests and assumptions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

Before you start

Recall Effect size and practical importance. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of statistical tests and assumptions. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of statistical tests and assumptions.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Effect size and practical importance. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sampling variability. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Baselines and fair comparisons. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of statistical tests and assumptions

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Effect size and practical importance. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain statistical tests and assumptions using the records and boundaries of this fictional investigation.

Notice the evidence

> Investigate the supplied Statistical tests and assumptions case. Separate observations, given facts and assumptions before choosing an explanation.

A statistical test compares observed **evidence** with a specified **null model**. A **p-value** is not the probability that the null hypothesis is true.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Statistical tests and assumptions. Connect each labelled input, deciding step and result.

Toy paired sign-flip test assumes independent paired differences whose signs are exchangeable under a symmetric zero-effect null. Enumerate all2^n sign assignments, holding magnitudes fixed.

Two-sided p is fraction with absolute mean at least the observed absolute mean; use inclusive ties.

Statistical tests and assumptions — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Three independent paired differences are 1, 1, 1. Under the stated symmetric zero-effect null, all eight independent sign assignments are equally likely. Use the absolute mean, include equally extreme ties, and compare the exact two-sided p-value with alpha 0.05.	Toy paired sign-flip test assumes independent paired differences whose signs are exchangeable under a symmetric zero-effect null. Enumerate all2^n sign assignments, holding magnitudes fixed. Two-sided p is fraction with absolute mean at least the observed absolute mean; use inclusive ties.	Only all-plus and all-minus have absolute mean1; others have1/3. Exact p=2/8=.25. At alpha .05 do not reject; this does not prove zero effect.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Statistical tests and assumptions. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Three independent paired differences are 1, 1, 1. Under the stated symmetric zero-effect null, all eight independent sign assignments are equally likely. Use the absolute mean, include equally extreme ties, and compare the exact two-sided p-value with alpha 0.05. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

> **Change one thing**
>
> Change: Break the task pairing and use unrelated A and B measurements.

> **Predicted result**
>
> **Prediction**: This paired sign-flip calculation is no longer justified by the stated design.

Why the result follows

Reason: Its differences must come from the same task, with the declared independent sign-symmetry assumptions.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Statistical tests and assumptions to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Null model	The probability model used for a test comparison
P-value	The null-model chance of an outcome at least as extreme under the defined statistic
Exchangeability	An assumption permitting the declared rearrangements

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Statistical tests and assumptions, predict a step, then compare with its explanation.

Three independent paired differences are 1, 1, 1. Under the stated symmetric zero-effect null, all eight independent sign assignments are equally likely. Use the absolute mean, include equally extreme ties, and compare the exact two-sided p-value with alpha 0.05.

Only all-plus and all-minus have absolute mean1; others have1/3. Exact p=2/8=.25. At alpha .05 do not reject; this does not prove zero effect.

Worked explanation · Statistical tests and assumptions

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Worked explanation · Statistical tests and assumptions

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
from itertools import product

def sign_flip_p(differences):
    """Two-sided exact test under independent symmetric sign exchangeability.

    Tiny n only; hold absolute magnitudes fixed and include equal extremes.
```

```
    A p-value is not a posterior probability that the null is true.
    """
    if not differences or len(differences) > 16:
        raise ValueError('require 1–16 paired units')
    threshold = abs(sum(differences))
    count = 0
    for signs in product((-1, 1), repeat=len(differences)):
        simulated = abs(sum((s * abs(d) for (s, d) in zip(signs, differences))))
        if simulated >= threshold - 1e-12:
            count += 1
    return count / 2 ** len(differences)

print(sign_flip_p([1, 1, 1]))
```

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Statistical tests and assumptions

A small exact paired sign-flip test. Compare three independent task pairs under a null that permits each difference’s sign to flip independently with equal probability.

The facts we will use

Supplied facts

A times [5,8,4] seconds; B [4,6,3]. Differences B–A are [-1,-2,-1]. Use a two-sided statistic: absolute sum of differences. Assume the stated sign-symmetry null is appropriate.

AI ACADEMY / AI-7/14

Three independent paired task times

	A seconds	B seconds	B-A seconds
Task 1	5	4	-1
Task 2	8	6	-2
Task 3	4	3	-1

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/14

Statistical tests and assumptions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Preserve paired differences

Differences are -1,-2,-1; their sum is -4 and mean is -4/3 seconds.



2. Enumerate null sign flips

Eight equally likely flips give sums -4,-2,0,2,4 with counts 1,2,2,2,1.



3. Count equally extreme sums

Only -4 and 4 have absolute sum at least 4: exact two-sided p=2/8=0.25.

CHANGE ONE THING

Break the task pairing and use unrelated A and B measurements.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Preserve paired differences	Differences are -1,-2,-1; their sum is -4 and mean is -4/3 seconds.
2. Enumerate null sign flips	Eight equally likely flips give sums -4,-2,0,2,4 with counts 1,2,2,2,1.
3. Count equally extreme sums	Only -4 and 4 have absolute sum at least 4: exact two-sided p=2/8=0.25.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A test compares an observed statistic with outcomes under a specified **null model**. Here the sign-flip assumptions make eight arrangements equally likely.

At a predeclared 0.05 threshold, p=0.25 does not reject the null; it does not prove no difference.

What this example does not establish

Keep the stated limit

Three pairs provide coarse **evidence**. The **p-value** is not the probability that the null is true;

invalid independence or symmetry assumptions invalidate this test.

Keep the stated limit

Read the labelled picture: Calculate each paired difference and the observed sum. Count null arrangements with absolute sum at least four, compute p, and explain the decision at threshold 0.05.

Keep the stated limit

Change one thing: Break the task pairing and use unrelated A and B measurements. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Statistical tests and assumptions claim and explain a repair using the supplied evidence.

The test requires its sign-**exchangeability** and independence assumptions; do not apply it blindly to clustered or asymmetric differences.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: The test requires its sign-**exchangeability** and independence assumptions; do not apply it blindly to clustered or asymmetric differences.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Statistical tests and assumptions with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Statistical tests and assumptions”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Statistical tests and assumptions. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Calculate each paired difference and the observed sum. Count null arrangements with absolute sum at least four, compute p, and explain the decision at threshold 0.05.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Statistical tests and assumptions. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Three independent paired differences are 1, 1, 1. Under the stated symmetric zero-effect null, all eight independent sign assignments are equally likely. Use the absolute mean, include equally extreme ties, and compare the exact two-sided p-value with alpha 0.05. Show the intermediate steps and give the conclusion with its scope.

```
from itertools import product

def sign_flip_p(differences):
    """Two-sided exact test under independent symmetric sign exchangeability.

    Tiny n only; hold absolute magnitudes fixed and include equal extremes.
    A p-value is not a posterior probability that the null is true.
    """
    if not differences or len(differences) > 16:
        raise ValueError('require 1–16 paired units')
    threshold = abs(sum(differences))
    count = 0
    for signs in product((-1, 1), repeat=len(differences)):
        simulated = abs(sum((s * abs(d) for (s, d) in zip(signs, differences))))
```

```
        if simulated >= threshold - 1e-12:
            count += 1
    return count / 2 ** len(differences)

print(sign_flip_p([1, 1, 1]))
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Statistical tests and assumptions. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: The test requires its sign-**exchangeability** and independence assumptions; do not apply it blindly to clustered or asymmetric differences.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

Add a labelled statistical tests and assumptions **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of statistical tests and assumptions.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Reproducible AI Research Study, save paired differences, the null assumptions, all eight sign arrangements and the decision rule.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **My limited claim and a useful next check**
>
> ---
>
> ---
>
> ---
>
> ---

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Statistical tests and assumptions case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Statistical tests and assumptions — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Three independent paired differences are 1, 1, 1. Under the stated symmetric zero-effect null, all eight independent sign assignments are equally likely. Use the absolute mean, include equally extreme ties, and compare the exact two-sided p-value with alpha 0.05.	Toy paired sign-flip test assumes independent paired differences whose signs are exchangeable under a symmetric zero-effect null. Enumerate all2^n sign assignments, holding magnitudes fixed. Two-sided p is fraction with absolute mean at least the observed absolute mean; use inclusive ties.	Only all-plus and all-minus have absolute mean1; others have1/3. Exact p=2/8=.25. At alpha .05 do not reject; this does not prove zero effect.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Three independent paired differences are 2, 2, 2. Assume a symmetric zero-effect null under which each sign flips independently with equal probability. Enumerate all eight assignments. The statistic is absolute mean; include ties as equally extreme. Compute the two-sided exact p-value and interpret at alpha 0.05.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Three independent paired differences are 2, 2, 2. Assume a symmetric zero-effect null under which each sign flips independently with equal probability. Enumerate all eight assignments. The statistic is absolute mean; include ties as equally extreme. Compute the two-sided exact p-value and interpret at alpha 0.05.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Statistical tests and assumptions, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A small exact paired sign-flip test. Compare three independent task pairs under a null that permits each difference's sign to flip independently with equal probability.

By the end, I can	How I will show it
Explain the mechanism and terminology of statistical tests and assumptions	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Statistical tests and assumptions without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of statistical tests and assumptions

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A small exact paired sign-flip test. Compare three independent task pairs under a null that permits each difference's sign to flip independently with equal probability.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Only -4 and 4 have absolute sum at least 4: exact two-sided p=2/8=0.25.

1. Enumerate null sign flips

2. Count equally extreme sums

3. Preserve paired differences

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Count equally extreme sums Only -4 and 4 have absolute sum at least 4: exact two-sided p=2/8=0.25.

A test compares an observed statistic with outcomes under a specified **null model**. Here the sign-flip assumptions make eight arrangements equally likely.

At a predeclared 0.05 threshold, p=0.25 does not reject the null; it does not prove no difference.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Break the task pairing and use unrelated A and B measurements. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This paired sign-flip calculation is no longer justified by the stated design.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The

answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: This paired sign-flip calculation is no longer justified by the stated design.

Its differences must come from the same task, with the declared independent sign-symmetry assumptions.

Repair a missing building block

Review Effect size and practical importance (ai-7/13). Goal: Explain the **mechanism** and terminology of effect size and practical importance. Short practice: Calculate the absolute error-rate reduction from 20/1000 to 15/1000.

Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?. Return to this week and explain what you changed.

Review Sampling variability (ai-7/11). Goal: Explain the **mechanism** and terminology of sampling variability. Short practice: Calculate the two four-case rates and their difference.

Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.. Return to this week and explain what you changed.

Review Baselines and fair comparisons (ai-7/8). Goal: Explain the **mechanism** and terminology of baselines and fair comparisons. Short practice: Calculate 6/8 and 7/8 correctness and name the one-case gain.

Why is giving B the answer key incompatible with an unseen comparison to the simple baseline?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

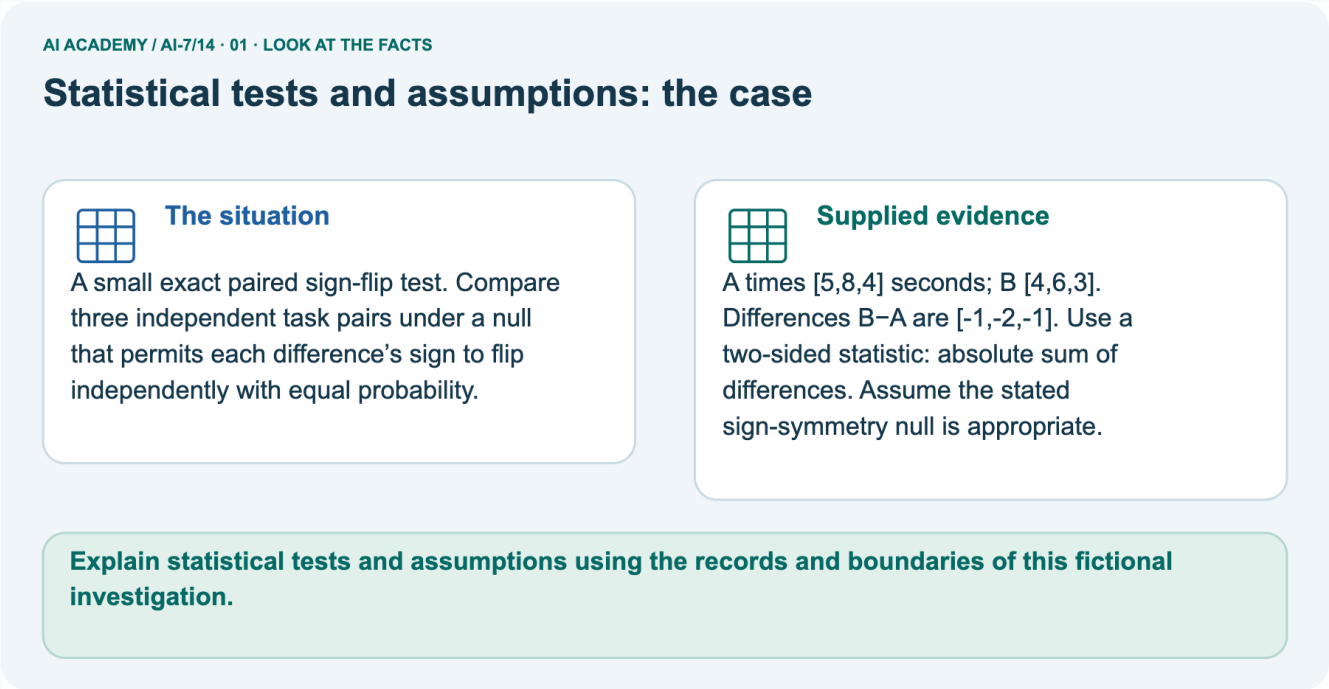
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Statistical tests and assumptions graphic using its labels, deciding evidence and relevant limitation.

01. Statistical tests and assumptions: the case



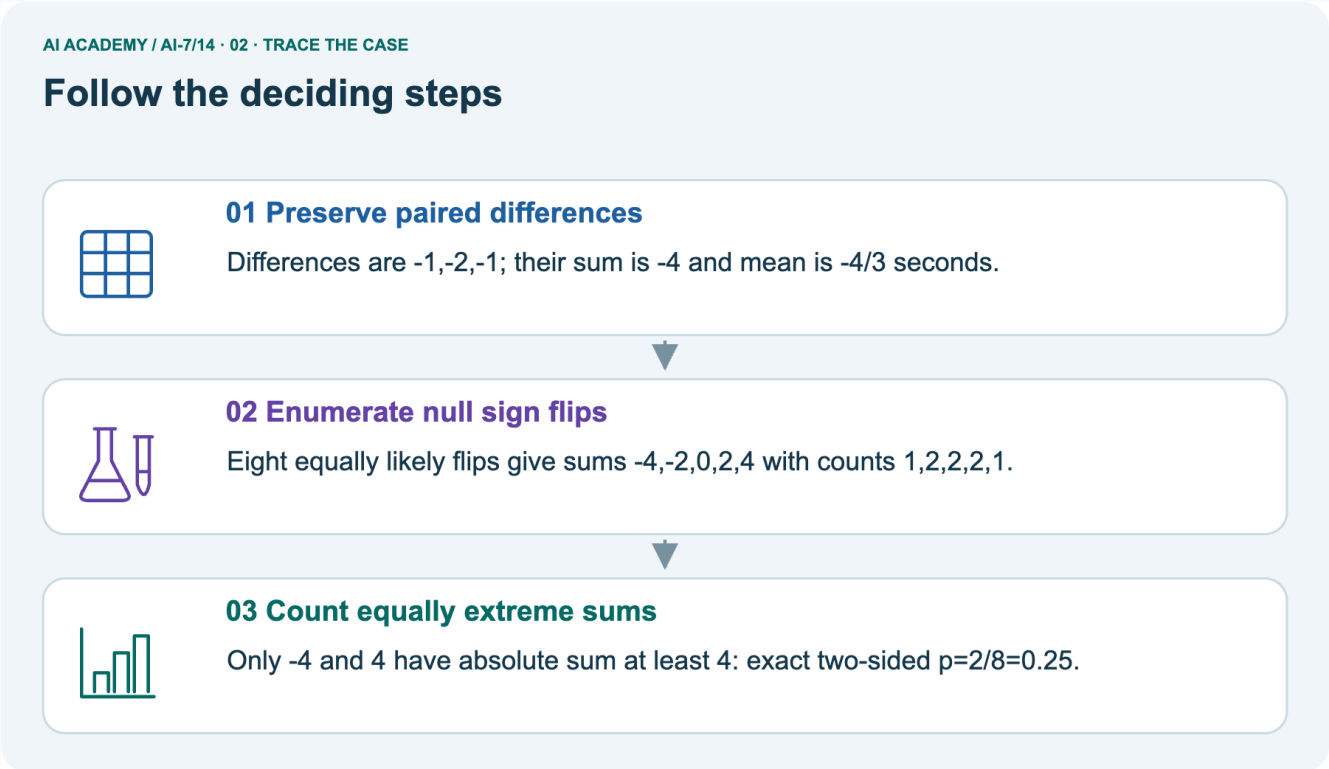
Follow the labelled picture with the matching explanation. Open the vector diagram

A small exact paired sign-flip test. Compare three independent task pairs under a null that permits each difference's sign to flip independently with equal probability. Supplied facts: A times [5,8,4] seconds; B [4,6,3].

Differences B–A are [-1,-2,-1]. Use a two-sided statistic: absolute sum of differences. Assume the stated sign-symmetry null is appropriate.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

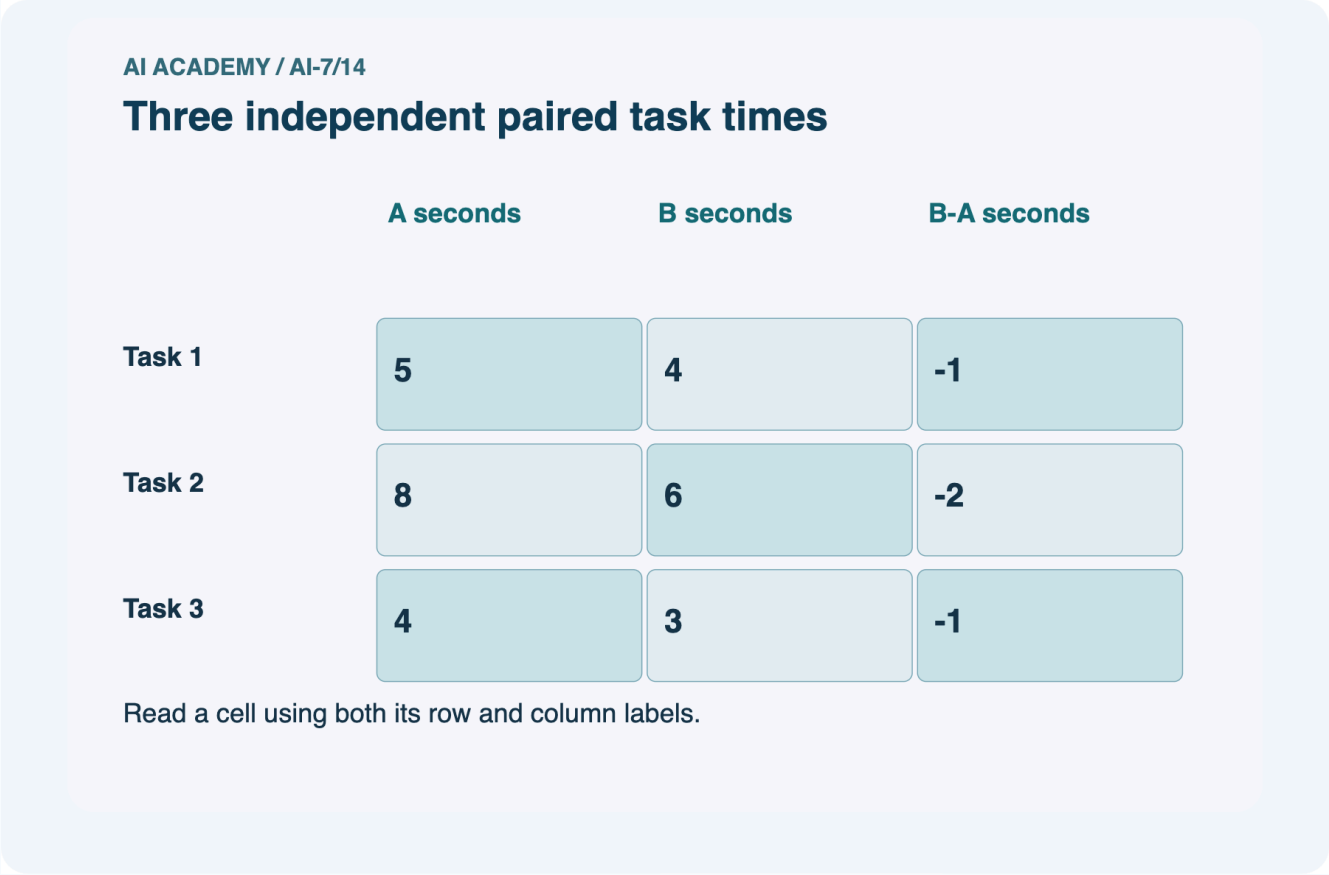


Follow the labelled picture with the matching explanation. Open the vector diagram

Preserve paired differences: Differences are -1,-2,-1; their sum is -4 and mean is -4/3 seconds. → Enumerate null sign flips: Eight equally likely flips give sums -4,-2,0,2,4 with counts 1,2,2,2,1. → Count equally extreme sums: Only -4 and 4 have absolute sum at least 4: exact two-sided p=2/8=0.25.

Point to each numbered stage and explain which recorded **input** it uses.

03. Three independent paired task times

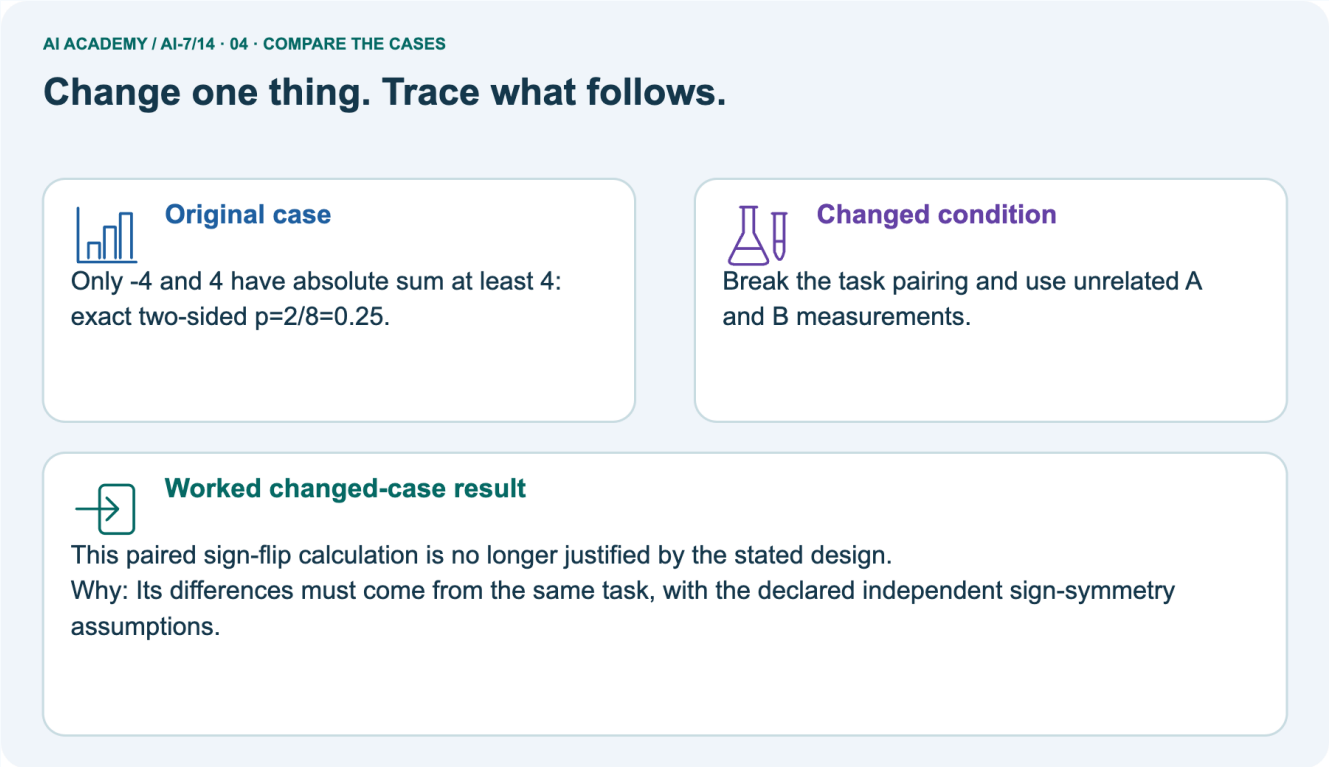


Follow the labelled picture with the matching explanation. Open the vector diagram

Three independent paired task times. Task 1: A seconds = 5, B seconds = 4, B–A seconds = -1. Task 2: A seconds = 8, B seconds = 6, B–A seconds = -2. Task 3: A seconds = 4, B seconds = 3, B–A seconds = -1.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



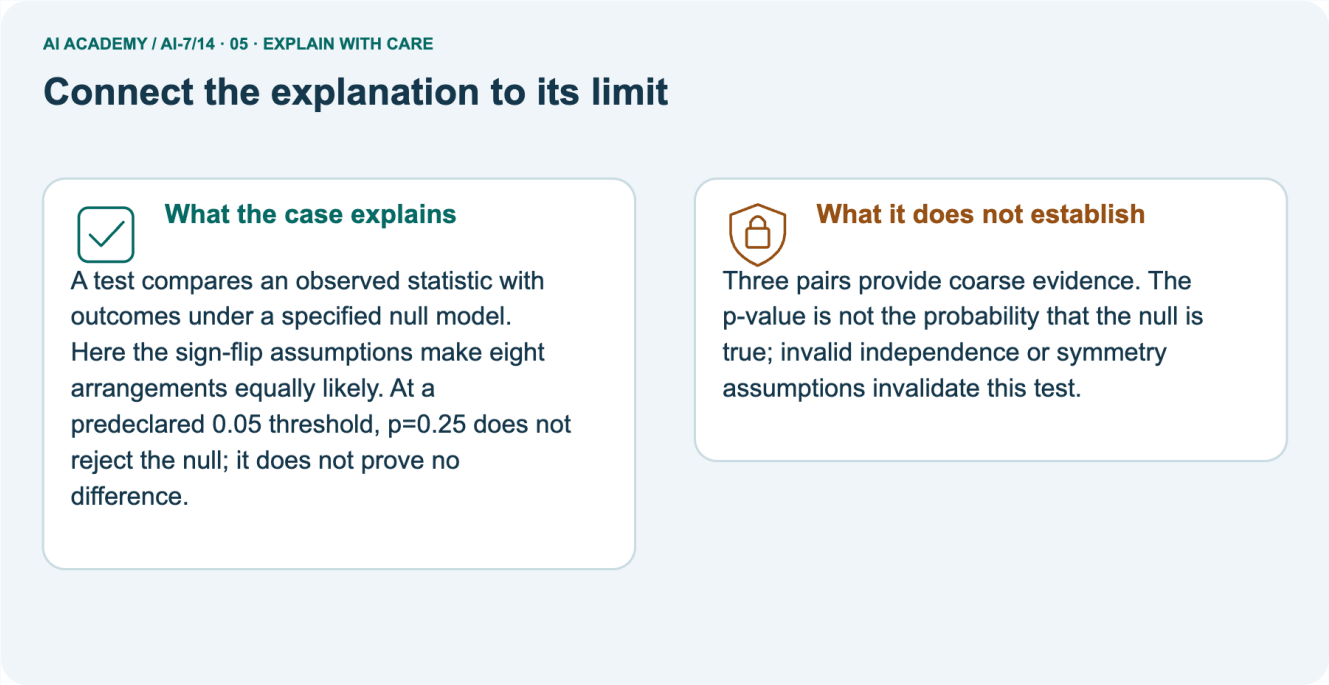
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Only -4 and 4 have absolute sum at least 4: exact two-sided p=2/8=0.25. Changed condition: Break the task pairing and use unrelated A and B measurements.

Expected result: This paired sign-flip calculation is no longer justified by the stated design. Reason: Its differences must come from the same task, with the declared independent sign-symmetry assumptions.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

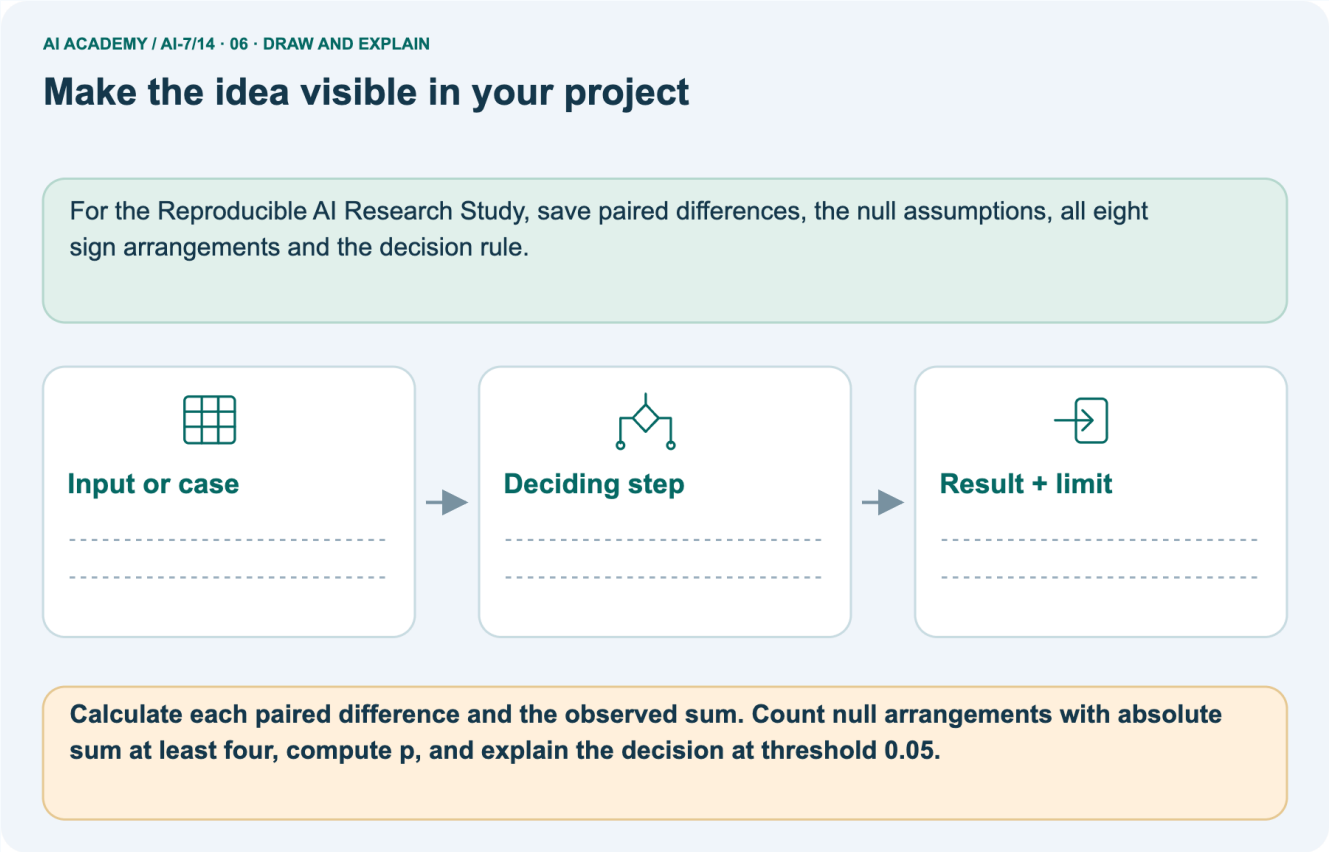
A test compares an observed statistic with outcomes under a specified **null model**. Here the sign-flip assumptions make eight arrangements equally likely.

At a predeclared 0.05 threshold, p=0.25 does not reject the null; it does not prove no difference. Limit: Three pairs provide coarse **evidence**.

The **p-value** is not the probability that the null is true; invalid independence or symmetry assumptions invalidate this test.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, save paired differences, the null assumptions, all eight sign arrangements and the decision rule. Draw your **input**, deciding step, result and limit.

Supported question: Calculate each paired difference and the observed sum. Count null arrangements with absolute sum at least four, compute p, and explain the decision at threshold 0.05.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.